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United States Patent Application Publication

20250283840

Kind Code

A1

Publication Date

September 11, 2025

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ALPHA DIFFRACTOMETER

Abstract

A diffractometer system includes an X-ray beam projector that projects an X-ray micro-beam at an analysis target, an X-ray receiver including an X-ray detector array to detect the transmitted X-ray beam passed through the object and X-rays that the target diffracts, and a computer workstation for system control and data analysis. The X-ray beam projector may include a radiation source, a beam forming system including at least one of a monochromator, a collimator, and focusing device. The computer workstation may control the X-ray devices and positioning mechanisms and motors, may acquire, process, store, or display data received from diffractometric examination, and may also calculate parameters of the three-dimensional reciprocal lattice of the analyzed target.

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Appl. No.: 19/217959

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18137356 20230420 parent-grant-document US 12332192 child US 19217959

us-provisional-application US 63333087 20220420

Publication Classification

Int. Cl.: G01N23/207 (20180101); G01N23/20008 (20180101)

Background/Summary

REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/137,356, filed on Apr. 20, 2023, which claims priority to U.S. Provisional Patent Application No. 63/333,087, filed Apr. 20, 2022, all of which is hereby incorporated by reference in their entirety.

BACKGROUND

[0002] X-ray analysis is widely used in science, technology, medicine, and veterinary medicine. Some advantages of X-ray analysis in medicine and veterinary medicine are that X-ray imaging is accessible, painless, and able produce detailed images of internal bones and soft tissues. More generally, radiological imaging techniques, including X-rays, CT scans, PET scans, MRIs, and ultrasonograms, may be the most informative method available for examining musculoskeletal systems, internal organs, neurology, ear nose and throat (ENT) structures, oral cavities, and diseases affecting such systems. Radiography is particularly important for the study of internal organs to detect foreign bodies and neoplasms, including malignant ones.

[0003] Small-angle X-ray spectroscopy has been proposed for study the structure of potential cancer sites. See, P. Lazarev et al., “Human Tissue X-ray Diffraction: Breast, Brain, and Prostate”, Proceedings of the 22nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (Cat. No. 00CH37143) Vol. 4, pp. 3230-3233 July 2000. The authors of this paper report that upon the incidence of X-ray radiation onto a sample, part of the radiation goes through the object unaffected, a part is absorbed, and a part is scattered. Absorption and incoherent scattering are related to the atomic composition and density of the sample. Coherent scattering is related to the molecular structure of the sample. The intensity of coherently scattered (or diffracted) radiation as a function of angle depends on the molecular structure of the object being irradiated, and every molecular structure has its own “signature” scattering pattern. Therefore, the study of scattering patterns of various materials can characterize molecular structures and recognize or distinguish specific molecular structures when encountered.

[0004] X-ray diffractometers that provide higher resolution measurements of scattered radiation can more effectively distinguish molecular structures, and several techniques may increase the spatial resolution of an X-ray diffractometer. One technique uses parallel or collimated X-ray micro-beams. For example, a parallel X-ray micro-beam may have a size of about $7 \times 5 \mu\text{m}$ with a small angular divergence, which aims high resolution strain measurements in a very local area. See, Tsusaka, Yoshiyuki; Yokoyama, Kazushi; et. al., “Formation of parallel X-ray micro-beam and its application”, Japanese Journal of Applied Physics, Part 2: Letters (2000), 39 (6B), L635-L637. The micro-beam was formed by compressing and collimating the X-rays from a 3rd-generation undulator source using successive asymmetrical reflections. Rocking curve measurements have used such beams to evaluate the strain that field oxidation induces in silicon (Si) wafers. In such cases, the lattice constant in the silicon region near the oxide film edge is contracted and in the SiO/Si transition region is extended. The difference between these lattice constants is as small as about $L1d/d \approx \pm 5 \times 10^{-6}$ but may be measured using X-ray diffraction techniques.

[0005] Another way to increase spatial resolution is to use a pixel detector with small pixel sizes, e.g., from 100-300 microns to 55 microns and below. See, N. Wermes, “Pixel Detectors”, arXiv: physics/0512037v1 [physics.ins-det] 5 Dec. 2005, Physikalisches Institut der Universitt Bonn, Germany; J. Treis, a, b, 1 R. Andritschke, et. al., “Pixel Detectors for X-ray Imaging Spectroscopy in

Space”, published by IOP publishing for Sissa, Pixel 2008 International Workshop Fermilab, Batavia, IL, U.S.A. 23-26 Sep. 2008. But reducing the detector pixel size may increase charge-shared events across pixels, which leads to a degradation of the image quality and energy resolution.

[0006] E. N. Gimenez et al., “Evaluation of the radiation hardness and Charge Summing Mode of a Medipix3-based detector with synchrotron radiation,” IEEE Nuclear Science Symposium & Medical Imaging Conference, Knoxville, TN, USA, 2010, pp. 1976-1980 notes that synchrotron applications such as coherent X-ray diffraction and X-ray photon-correlation spectroscopy require detectors with a pixel pitch of 50 μm as highlighted by a survey with beamline scientists of Diamond Light Source synchrotron. Furthermore, an X-ray detector should also have a high frame rate, large dynamic range, and large detection efficiency. The Medipix3 readout chip with a pixel pitch of 55 μm emerged as a good candidate to develop a new detector for the aforementioned applications. Additionally, the Medipix3 readout chip implements an operating mode, referred to as Charge Summing Mode (CSM), with the purpose of eliminating charge-shared events. This mode can be very useful since the charge-sharing effect increases as the detector pixel size decreases. Also, the Medipix3 readout chip may be more radiation hard than its predecessor Medipix2.

[0007] U.S. Pat. No. 6,175,117, entitled “Tissue analysis apparatus” describes an X-ray apparatus that measures small-angle X-rays for analysis of substances within a breast. The apparatus includes a breast positioning area, a beam forming apparatus that forms X-rays into at least one beam, and a detector that detects a scattering pattern of X-rays scattered from the portion of the breast positioned within the breast positioning area. The X-ray apparatus uses a filter as a stopper of the incident beam to prevent the transmitted X-rays from harming (e.g., overheating or destroying) the detector. The filter has an upper surface with a shape matching the shape of the cross section of the transmitted beam and must be accurately positioned in the transmitted beam such that the upper surface is matched to the outer edges of the transmitted beam. During X-ray diffraction analysis, the filter needs to be moved in and out of the transmitted beam as a detector position adjuster moves the detector relative to the positioning area. All this places high demands on the accuracy of manufacturing the structural elements of the apparatus and makes the apparatus complex to use.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 shows perspective view of a diffractometer in accordance with an example of the present disclosure.

[0009] FIG. 2 is a block diagram of a diffractometer in accordance with an example of the present disclosure.

[0010] FIG. 3 shows a schematic view of the detector stage in accordance with an example of the present disclosure capable of moving an X-ray detector array along the direction of an X-ray beam and to oscillate the X-ray detector array in a plane perpendicular to the X-ray beam.

[0011] FIG. 4 shows an example of a diffractometer system in which an X-ray transmitter and an X-ray receiver are mounted on separate telescoping arms.

[0012] FIG. 5 shows a perspective view of a C-arm diffractometer system in accordance with an example of the present disclosure.

[0013] FIG. 6 shows a top view of a C-arm diffractometer system in accordance with an example of the present disclosure.

DETAILED DESCRIPTION

[0014] An X-ray diffractometer in accordance with some examples disclosed herein may perform X-ray structural analysis of objects including parts of living humans or other animals, biological tissue samples, and physical and chemical samples. Some examples of the x-ray diffractometer

allow 3-D diffractometric structural analysis, as well as calculation of the parameters of the three-dimensional reciprocal (reverse) lattice of an object under study. In particular, an X-ray diffractometer in accordance with an example disclosed herein can detect cancerous areas in the internal organs of humans and animals and study changes in the structure of the tissue in areas.

[0015] An X-ray diffractometer in accordance with one example of the present disclosure does not employ a beam stopper and thereby overcomes drawbacks of prior diffractometric devices, such as technical difficulties of fabrication, complexity of configuration, and low resolution. In addition, a diffractometer such as disclosed herein has resistance to the external environment and insensitivity to external mechanical influences. Another feature of some examples of the disclosed diffractometer is the ability of a two-dimensional pixel detector to mechanically move along the direction of the incident X-ray beam and to perform oscillatory movements in a plane perpendicular to the incident beam.

[0016] An X-ray diffractometer in accordance with a specific example includes a computer workstation, an X-ray projector directing an incident micro-beam of X-rays at the analyzed object in a positioning area, and a receiver including an X-ray detector array capable of detecting the transmitted X-ray micro-beam passed through the analyzed object and detecting all or a portion of the X-rays that the analyzed object diffracts or otherwise scatters. The X-ray beam projector may include a radiation source operating in continuous mode and an apparatus forming X-ray micro-beam. The beam forming apparatus may include at least one monochromator and at least one collimating and focusing optical device. The computer workstation may be configured to control the X-ray transmitter and receiver, to control positioning mechanisms and motors and to process, store, and display data from 3-D diffractometric examinations and structural analysis. The workstation may also be configured to calculate parameters of a three-dimensional reciprocal lattice in the analyzed object.

[0017] FIG. 1 shows a perspective view of an X-ray diffractometer **100** in accordance with an example of the present disclosure, and FIG. 2 is a block diagram identifying in more detail some components of X-ray diffractometer **100**. As shown in FIGS. 1 and 2, X-ray diffractometer **100** includes a transmitter housing **102** containing an X-ray beam projector **110** and a receiver housing **104** containing an X-ray detector **120**. Transmitter housing **102** and receiver housing **104** are separated from each other by (or otherwise define) an intervening examination volume **130** in which a target **132** to be examined may be held between X-ray beam projector **110** and X-ray detector **120**.

[0018] In one specific example, transmitter housing **102** or receiver housing **104** is a protective container that holds one or more X-ray devices, e.g., projector **110** or detector **120**, in a vacuum or surrounded by an inert gas such as helium or neon. Each of housings **102** and **104** has a side wall **103** or **105** facing target **132**, and all or a portion of each side wall **103** or **105** may be transparent to X-rays. For example, wall **103** may include a beryllium window through which X-rays from projector **110** enter examination volume **130**, and wall **105** may include another beryllium window through which X-rays pass from examination volume **130** to X-ray detector **120** inside receiver housing **104**. Each container **102** or **104** may specifically be filled with an inert gas such as neon or helium. In yet another example, diffractometer **100** further employs a chamber filled with an inert gas which is placed between detector **120** and target **132** in a working state and is set aside in non-working state.

[0019] X-ray beam projector **110** directs an X-ray beam **134** into target **132** in examination volume **130**. A portion of X-ray beam **134** may be transmitted through target **132** and forms a transmitted X-ray beam **136** that may be parallel or colinear with incident X-ray beam **134**. Another portion of incident X-ray beam **134** may interact with target **132** producing scattered X-rays **138**, which may include coherently scattered X-rays, i.e., diffracted X-rays or incoherently scattered X-rays.

[0020] X-ray projector **110** in the example of FIG. 2 includes an X-ray source **112** and beam forming system **114**. X-ray source **112** may, for example, be an X-ray tube or X-ray laser. X-ray

micro-beam forming system **114** may include one or more of an X-ray collimator **115**, an X-ray focusing element, and a filter **117**. X-ray collimator **115** may, for example, include a Kratki or Montel mirror collimator. Alternatively, X-ray collimator **115** may include a pair of X-ray mirrors that are arranged orthogonally relative to each other where each X-ray mirror is a multilayer film mirror. Focusing systems **116** may include grazing-incidence mirrors that are slightly curved to focus X-ray beam **134**. Filter systems **117** of x-ray beam forming system **114** may include, for example, at least one monochromator. In one specific example, x-ray source **112** includes a radiation source operating in continuous mode, e.g., x-ray tube, to produce monochromatic x-rays and x-ray beam conditioner **114** includes a collimating mirror and aperture that form or shape the x-ray micro-beam **112**. Alternatively, x-ray beam conditioner **114** may employ any other systems capable of controlling the width, collimation, or other characteristics of x-ray beam **142**.

[0021] In operation, beam conditioner **114** receives x-rays from x-ray source **112** and produces an x-ray beam **142** having desired intensity, wavelength, and beam profile. X-ray beam **142** may, for example, be is a micro-beam having a FWHM of about 200 microns or less, a wavelength of 1.60 Å to 0.3 Å, and an intensity on the order of about 10¹² photons/sec/mm.^{sup.2}.

[0022] In some examples disclosed herein, a visible light laser **118** may be mounted in or on transmitter housing **102**, and an optical system may direct a beam **119** of visible along a path parallel to X-ray micro-beam **134**. In examination volume **130**, visible beam **119** may coincide with or propagate adjacent to X-ray micro-beam **134**, so that beam **119** may be used to point X-ray beam **134** at selected points on the target **132** being examined.

[0023] X-ray detector **120** includes an X-ray detector array **122** on a mounting system or stage **124**. In one example disclosed herein, X-ray detector array **122** is an X-ray detector array including a two-dimensional array of detector elements, each of which provides a signal upon receiving of X-ray photon. Stage **124** generally holds detector array **122** so that transmitted X-ray beam **146** is perpendicular to the surface of detector array **122**. Stage **124** is also capable of moving detector array **122**. More particularly, stage **124** includes a mechanism **126** to mechanically move detector array **122** toward or away from target **132** along the propagation direction of transmitted beam **136** and a mechanism **128** to transversely move or oscillate detector array **122** in a plane perpendicular to the propagation direction of transmitted beam **136**. As described further below, the oscillation capabilities of stage **134** allow use of diffractometer **100** without a beam stopper to protect X-ray detector array **122**, while still avoiding damage that the intensity of transmitted X-ray beam **136** might otherwise cause to X-ray detector array **122**. The absence of a beam stopper avoids parasitic scattering from the beam stopper and reduces noise in measurements of scattered X-rays **138**.

[0024] Stage **124** (particularly mechanism **126**) as described above can move detector array **122** relative to target **132** along the propagation direction of X-ray beams **134** and **136**. For movement along the beam direction, mechanism **126** may include a drive train similar to the drive trains used to adjust the height of optics assemblies in photocopiers. Movement of detector array **122** relative to target **132** changes the angular resolution and the angular range of detector array **122** for measurements of scattered radiation **138**. For instance, moving the detector further from target **132** increases the area of detector array **122** exposed to radiation scattered over a smaller range of angles. On the other hand, moving the detector closer to the object increases the area of detector array **122** exposed to radiation scattered over a larger range of angles, e.g., up to close to 90°. The larger the detector area exposed to radiation scattered over a fixed range of angles, the higher the resolution of the detector with respect to those angles. Mechanism **124** may provide a range of movement that produces approximately the same resolution of diffraction measurements at small (less than 1°) and large (approximately about 90°) of diffraction angles. Stage **124** may thus be used to position detector array **122** to achieve a desired resolution when measuring radiation **138** scattered to angles of interest.

[0025] Incident X-ray beam **134** may be a collimated beam, e.g., contain X-rays having a uniform direction of travel, or may be focused, e.g., be converging on a focal point at or near the active

surface of detector array **122** to effectively increase the angular range of scattered X-rays that detector array **122** may measure. Transmitted x-ray beam **136** may have similar beam characteristics, e.g., size, collimation, or convergence, as incident X-ray beam **134**. In some implementations, the area of detector array **122** receiving transmitted X-ray beam **136** may be less than size of a single pixel on detector array **122**, which allows a single pixel at time to capture and measure transmitted beam **136**, while the rest of the area of detector array **122** detects scattered X-rays **138**.

[0026] FIG. 2 illustrates an example of X-ray diffractometer **100** in which stage **124** includes a mechanism **128** that excites transverse, i.e., transverse to the direction of transmitted X-ray beam **136**, vibrations of two-dimensional detector array **122**. Several types of transverse vibrations may be employed. For example, one type of vibration may include linear fluctuations only in the equatorial (horizontal) direction. Alternatively, vibration may include linear fluctuations only in the meridional (vertical) direction. Yet another type of vibration may include fluctuations in the equatorial and meridional directions simultaneously. The transverse vibrations of X-ray detector array **122** may be due to the excitement of a longitudinal surface acoustic wave. In one example, mechanism **128** uses a piezoelectric effect and moves detector array **122** in response to one or more oscillating electrical signals. To avoid having transmitted X-ray beam **136** constantly incident on the same pixel or pixels of detector array **122**, the amplitude of the transverse vibrations of two-dimensional detector array **122** may be greater than or equal to a pitch of the pixels in detector array **122** and greater than or equal to the width of transmitted X-ray beam **136** at detector array **122**. The frequency of the transverse vibrations of the two-dimensional detector array **122** may be less than or equal to a frame rate at which detector array **122** captures measurements of transmitted and scattered X-rays **136** and **138**. In one example, a readout chip **125**, which may control the frame rate for two-dimensional detector array **122**, implements an operating mode referred to as Charge Summing Mode (CSM) with the purpose of eliminating charge-shared events.

[0027] A diffractometer as disclosed herein may move X-ray projector **110** and X-ray detector **120** to obtain 3D-diffractometric X-ray diffraction data without changing object position, e.g., the position of target **132**. Alternatively, target **132** may be moved, e.g., rotated around the vertical axis, so that diffractometer **200** can obtain the 3D-diffractometric X-ray diffraction data. In another example disclosed herein, the computer workstation executes process controlling programs that perform 3-D diffractometric data collection at different angles of rotation of a C-arc in each mutually perpendicular planes sequentially, and for each angle of rotation of the C-arc in these planes, a series of diffraction measurements is carried out with a change in the distance from the analyzed subject to the two-dimensional pixel detector.

[0028] In still another example disclosed herein X-ray beam projector **110** and the receiver of X-rays are mounted on telescoping arms attached to a wall or a ceiling of the laboratory room, wherein these X-ray devices are located at one or two telescoping arms and furthermore the diffractometer further includes at least one housing including mechanisms and motors designed to move these telescoping arms.

[0029] In one example disclosed herein the detector pixels are characterized by different values of the maximum permissible power density of X-ray radiation incident on one pixel, moreover, the pixels located in the central region of the flat pixel-detector have the highest maximum permissible power density compared to the pixels located on the periphery. In another example disclosed herein the maximum permissible power density of each pixel is inversely proportional to the distance between this pixel and the central pixel of the pixel-detector.

[0030] In one example disclosed herein, the diffractometer further includes facilities and mechanisms designed to move the positioning area for the examination target between the X-ray beam projector and the receiver along the direction of the incident X-ray beam. By way of example, and not by way of limitation, as facilities and mechanisms may be used mechanical guides and stepper motors. In another example disclosed herein the diffractometer further includes

a rectification collimator located in the positioning area near the analyzed target **132** and facing X-ray beam projector **110**.

[0031] FIG. **3** shows a schematic view of an oscillating stage or mounting system **300** capable of moving an X-ray detector array along the direction of an incident (transmitted) X-ray beam and capable of oscillating the X-ray detector array in a plane perpendicular to the X-ray beam direction. Mounting system **300** includes a base **1** that is mounted on a guide **2**, which permits base **1** to be slid along the beam direction. For example, base **1** and guide **2** may be connected through a ball-screw motion transmission that transmits the rotational movement from an electrical motor **3** to the translational movement of base **1**.

[0032] Structure for transverse oscillation of the detector array may be mounted on base **1**. As shown in FIG. **3**, a structural element **4** is mounted in slots **5** in base **1**, which permits structural element **4** to move or oscillate in the vertical direction along vertical guides **5**. An exciter **6** is coupled between base **1** and structural element **4** to drive movement or oscillations of element **4** along guides **5**. Exciter **6** may, for example, include a stepper motor with coupling or transmission such that an electric signal applied to exciter **6** controls the vertical position of structural element **4** relative to base **1**. Another structural element **7** engages with horizontal guides **8** in structural element **4**, and an exciter **9** coupled to structural elements **4** and **7** drives movement or oscillations of structural element **7** back and forth along horizontal guides **8**. Exciters **8** may include a stepper motor with coupling or transmission such that an electric signal applied to exciter **8** controls the horizontal position of structural element **7** relative to structural element **4**. A rod, pin, or other structure **10** may provide a mounting for the X-ray detector array or alternatively an X-ray projector.

[0033] In still another example disclosed herein, a diffractometer includes one or more devices that excite transverse vibrations of the X-ray beam projector, e.g., X-ray beam projector **110** of FIG. **1**, so that the incident X-ray micro-beam, e.g., X-ray beam **134**, moves on the analyzed object parallel to itself and the location where the transmitted X-ray micro-beam, e.g., beam **136**, strikes the detector array also move. For this example, mounting system **300** may hold the X-ray beam projector.

[0034] FIG. **4** shows an example of a diffractometer system **400** including an X-ray transmitter **402** and an X-ray receiver **404**, which may be similar or identical to the X-ray devices described above. In particular, X-ray transmitter **402** provides an incident X-ray micro-beam **134** directed at an analyzed target **132** as described above with reference to FIGS. **1** and **2**. (Walls of transmitter **402** and receiver **404** facing target **132** are transparent to X-rays.) In FIG. **4**, target **132** is on a rotary positioning device **430**. X-ray receiver **404** contains a movable two-dimensional X-ray detector array in a protective container that may be vacuumed or filled with an inert gas as described above. The X-ray detector array may be moved with the help of an oscillating mounting structure such as shown in FIG. **3**. The X-ray detector array may be designed to detect both the transmitted micro-beam of X-ray **136** passed through analyzed target **132** and all or part of the X-rays **138** that are diffracted or otherwise scattered by target **132**.

[0035] X-ray transmitter **402** and X-ray receiver **404** of diffractometer system **400** are respectively mounted on telescoping arms **18** and **19**, which may be attached to a ceiling of the laboratory or examination room. In particular, system **400** includes two housings **20** and **22** that contain mechanisms and motors designed to move the telescoping arms **18** and **19** along respective guides **21** and **23** that may be fixed on a ceiling. Housings **20** and **22** may also contain mechanisms and motors to extend or retract telescoping arms **18** and **19** to move X-ray transmitter **402** and X-ray receiver **404** vertically up or down and to rotate X-ray transmitter **402** and X-ray receiver **404** about axes of respective telescoping arms **18** and **19**. A characteristic feature of this example is the mechanical autonomy of the mechanisms of movement of X-ray transmitter **402** and X-ray receiver **404**. These elements of the diffractometer **400** may be interconnected exclusively through a workstation **24** via electrical interconnections particularly via wireless communications. In

diffractometer system **400**, the X-ray detector array simultaneously detects the transmitted micro-beam **136** of X-ray passed through the analyzed target **132** and all or a portion X-rays **138**. These features of diffractometer system **400** reduce the requirements on the accuracy of manufacturing of the mechanical structure and on the rigidity of the bearing elements and simplifies commissioning of diffractometer system **400**. All this leads to a reduction in the costs of diffractometer system **400**. [0036] Computer workstation **24** may be configured to and communicate with beam projector **402** and X-ray receiver **404** through wired or wireless communication links **25** and **26**. Computer workstation **24** may execute software or firmware for control of X-ray devices in X-ray transmitter **402** and X-ray receiver **404** and control of the mechanisms and motors in housings **20** and **22**. Workstation **240** may also execute processes for acquisition, processing, storing, and displaying of data received from the 3-D diffractometric structural examination, and also calculation of the parameters of the three-dimensional reciprocal lattice of the studied subject.

[0037] In still another example disclosed herein, an X-ray beam projector and an X-ray detector may be mounted on a C-arc or C-arm device **500** as shown in the perspective view of FIG. 5 and the top view of FIG. 6. In C-arm device **500**, transmitter and receiver housings **102** and **104** such as described above with reference to FIGS. 1 and 2 are at opposite ends a generally c-shaped structure or C-arm **510**. C-arm **510** is mounted on a support **522** extending from a drive system **520**. Drive system **520** further includes motors and mechanisms to move support **522** and C-arm **510** relative to a positioning area **532** on a holding structure **530** for the target of X-ray analysis, e.g., a breast. Drive system **520** may particularly move support **522** horizontally along a guide **524** on the surface of a base **30** as shown in FIG. 6 to take C-arm **510** to a working position in positioning area **50**. In addition, C-arm **510** can move perpendicular to base **30** along a guide **40**. To move the C-arm **510** to examination area **532** and back, by way of example, and not by way of limitation, a ball-screw motion transmission **560** may be used which transmits the rotational movement from an electrical motor **570** to the translational movement of C-arm **510**. To move second C-arm **510** perpendicular to the plane of base **30**, by way of example, and not by way of limitation, a ball-screw motion transmission **280** is used which transmits the rotational movement from an electrical motor **290** to the translational movement.

[0038] C-arm **510** may more generally be movable in three mutually perpendicular planes around a target in positioning area **532**. In one example, motors **570** and **590** can power movement C-arm **510** independently in the vertical and horizontal directions. C-arm **510** may be further movable to rotate around its symmetry axis and around an axis perpendicular to the C-arm plane. The movement of C-arm **510** may be realized using a parallel translation mechanism such as a ball-screw motion transmission or a combination of a motor-driven feed screw shaft with a slide member that is held in thread-engagement with the feed screw shaft. In another example, rotation of C-arm **510** may be realized by using an appropriate rotary drive mechanism such as a tangent bar type drive mechanism where the drive source is preferably a position-controllable electric motor such as a pulse motor or a servomotor.

[0039] Although aspects of the present disclosure have been described in detail with reference to certain implementations, persons possessing ordinary skill in the art to which this disclosure pertains will appreciate that various modifications and enhancements may be made without departing from the spirit and scope of the claims that follow. Any feature, whether preferred or not may be combined with any other feature whether preferred or not. It should be understood that various alternatives to the examples of the invention described herein can be employed in practicing the invention. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby. The appended claims are not to be interpreted as including means-plus-function limitations, unless such a limitation is explicitly recited in a given claim using the phrase “means for.”

Claims

1. A diffractometer comprising: an X-ray beam projector positioned and aligned to direct an X-ray beam at a target to be examined; and an X-ray receiver comprising an X-ray detector array positioned to detect the X-ray beam passed through the target; wherein the X-ray beam projector comprises a radiation source and an apparatus forming an X-ray micro-beam; wherein the X-ray detector array comprises pixels having different values of maximum permissible power density of X-ray radiation incident on one pixel; and wherein the maximum permissible power density of each pixel is inversely proportional to a distance between the pixel and a central pixel of the X-ray detector array.
2. The diffractometer of claim 1, wherein the X-ray detector array is in a protective container that is vacuumed or filled with an inert gas.
3. The diffractometer of claim 2, wherein the inert gas is neon or helium.
4. The diffractometer of claim 1, wherein the X-ray receiver comprises 1) a mechanism for moving the X-ray detector array along a direction of the X-ray micro-beam that provides approximately a same resolution of diffraction measurements at diffraction angles less than 1 degree and approximately about 90 degrees, and 2) a device designed for excitation of transverse vibrations of a two-dimensional detector array in relation to the X-ray beam.
5. The diffractometer of claim 4, wherein the device designed for excitation of transverse vibrations employs a piezoelectric effect.
6. The diffractometer of claim 4, wherein an amplitude of the transverse vibrations of the two-dimensional detector array is not less than a pixel's pitch.
7. The diffractometer of claim 4, wherein a frequency of the transverse vibrations of the two-dimensional detector array is not more than a frame rate.
8. The diffractometer of claim 4, wherein the transverse vibrations of the two-dimensional detector array are carried out due to the excitement of a longitudinal surface acoustic wave.
9. The diffractometer of claim 1, wherein the radiation source is selected from an X-ray tube or an X-ray laser.
10. The diffractometer of claim 1, wherein the X-ray beam projector and the X-ray receiver are located at opposite ends of a C-arc, and the diffractometer further comprises a housing containing mechanisms and motors that move and rotate the C-arc in three mutually perpendicular planes around the target.
11. The diffractometer of claim 10, further comprising: a computer; wherein the computer executes a process controlling program that performs 3-D diffractometric data collection at different angles of rotation of the C-arc sequentially, and for each angle of rotation of the C-arc, a series of diffraction measurements is carried out with a change in distance from the target to a two-dimensional detector array.
12. The diffractometer of claim 1, wherein the X-ray beam projector further includes a Kratki or Montel mirror collimator.
13. The diffractometer of claim 1, wherein the X-ray beam projector further comprises an X-ray collimating mirror with a pair of X-ray mirrors that are arranged orthogonally relative to each other, wherein the X-ray mirrors are multilayer film mirrors.
14. The diffractometer of claim 1, wherein the X-ray detector array includes a plurality of detector elements each of which provides a signal upon receiving an X-ray photon.
15. The diffractometer of claim 1, further comprising a laser of visible light configured to direct a visible beam to point the X-ray micro-beam at selected points on the target.
16. The diffractometer of claim 1, wherein the X-ray beam projector and the X-ray receiver are mounted on telescoping arms.
17. The diffractometer of claim 1, wherein the pixels of the X-ray detector array are located in a

central region of the X-ray detector array having a highest maximum permissible power density compared to the pixels located on a periphery of the X-ray detector array.

18. The diffractometer of claim 1, further comprising a device designed for excitation of transverse vibrations of the X-ray beam projector so that the X-ray micro-beam moves on the target parallel to itself.

19. The diffractometer of claim 1, wherein the X-ray beam projector and the X-ray receiver are mechanically autonomous elements and are interconnected by means of a computer via electrical interconnections including only wireless communications.

20. The diffractometer of claim 1, further comprising: a computer that is intended for controlling the X-ray beam projector, the X-ray receiver, and mechanisms and motors, and for processing, storing and displaying data received from 3-D diffractometric structural analysis, and also for calculation of parameters of a three-dimensional reciprocal lattice of the target to be examined; wherein: the radiation source operates in continuous mode; and the apparatus forming the X-ray micro-beam comprises a monochromator and a collimating and focusing optical device.
